Keisuke Yoshihara

Experimental Particle Physicist

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Research profile: Experimental particle physics at Belle II and SuperKEKB, spanning searches for physics beyond the Standard Model; operation and readout of the time-of-propagation (TOP) and K_L^0 and muon (KLM) detectors; FPGA firmware development; and open-source mixed-signal ASIC design. Current instrumentation work includes ultrafast beam diagnostics with the bunch oscillation recorder (BOR) and silicon-based X-ray radiation monitor (SiXRM).

Education

Doctor of Philosophy, Physics, The University of Tokyo, 04/2011 - 09/2014
Thesis: "Measurement of the Higgs boson couplings using the $WW^* \to \ell\nu\ell\nu$ final state." [Link]
Master of Science, Physics, The University of Tokyo, 04/2009 - 03/2011,
Thesis: "R&D studies of Muon Polarimeter for J-PARC E06 T-Violation Experiment." [Link]
Bachelor of Science, Physics, Yokohama National University, 04/2005 - 03/2009,
Thesis: "Calculation of higher order diagrams for Compton Scattering on High Energy Limit."

Professional Experience

Assistant Professor, University of Hawai'i at Mānoa, 04/2024 - present
Visiting Researcher, Nagoya University KMI, 05/2023 - present
Data Science Consultant, Accenture, 05/2023 - 03/2024
Visiting Associate Professor, IPNS KEK, 04/2022 - 04/2023
Designated Associate Professor, Nagoya University KMI, 04/2021 - 04/2023
Postdoctoral Fellow, Physics, Iowa State University, 10/2019 - 03/2021
Postdoctoral Fellow, Physics, University of Pennsylvania, 10/2014 - 09/2019

Awards

- ATLAS Outstanding Achievement Award, ATLAS Collaboration, 01/2021 [Link]
- Young Scientist Award of the Physical Society of Japan, JPS, 03/2016 [Link]
- Incentive Award of the Japan Association of High Energy Physicists, JAHEP, 10/2015 [Link]
- Graduation with the highest honors (GPA 3.92/4.00), Yokohama National University, 03/2009

Research Grants

- Enhanced X-ray Beam Size Monitor (XRM) to Improve SuperKEKB Stability and Luminosity
  (Co-PI, U.S. DOE U.S.-Japan Science and Technology Cooperation Program in High Energy Physics, $235,000) – awarded, 2026
- Construction of a Fast Beam Diagnosis System
  (PI, JSPS KAKENHI, 17.7 million JPY) – awarded, 04/2022 [Link]
- High Energy Physics Research at the University of Hawaii – Feasibility Study of an Open-Source Mixed-Signal ASIC
  (Co-PI, DOE FOA-3432, $75,000) – awarded, 2026

Leadership

- Belle II Professional Development Committee Member (01/2026-present)

- US Belle II DOE Level-2 Manager for TOP detector (07/2025-present)
- US Belle II Executive Committee Member (01/2025-present)
- Belle II Upgrade WG Coordinator of Outer Detector Sub-group, 09/2022 - 04/2023
- Belle II (deputy) Run Coordinator, 05/2020 - 04/2023
- Belle II KLM (deputy) Run Coordinator, 02/2020 - 03/2021
- ATLAS SUSY Background Forum Convener, 10/2017 - 09/2019
- ATLAS SUSY Stop1L Analysis Contact, 04/2016 - 10/2019
- ATLAS TRT DAQ Operation Leader, 04/2017 - 03/2019
- ATLAS SUSY Higgsino Disappearing Track Analysis Contact, 10/2017 - 1/2018
- ATLAS SUSY MC Generation Coordination, 11/2016 - 10/2017
- ATLAS SUSY Fake-lepton Review Forum Coordinator, 05/2016 - 04/2017

Mentoring at UH since 2024

- **Current Postdoctoral Researchers:** Tommy Lam, Horacio Crotte Ledesma, Martin Bessner, Shahab Kohani, Vasily Shebalin.
- **Former Postdoctoral Researchers:** Christopher Ketter, Harsh Purwar.
- **Graduate Students:** Cody Driver, Jacob Romeo, Robert Sinesi, Vikas Kumar.
- **Undergraduate Students:** Aleena Bendzius, Yeunggyun Kwon.

Organization experience

- ML4FE Workshop 2025 [Link], 05/2025
- 2nd KMI Flash [Link], 02/2023
- Flavor Physics workshop 2022 [Link], **a main organizer**, 11/2022
- 1st KMI Flash [Link], 07/2021

I also have experience organizing workshops and meetings, and serving as a session chair at international conferences.

Teaching

- "PHYS476: Modern Electronics for Physicists", UH Manoa, Spring 2025, Spring 2026.
  Project-based course covering digital circuits, FPGA programming, Verilog HDL, and AI implementation on FPGA hardware.
- "PHYS399: Introduction to Scientific Programming", UH Manoa, Fall 2025.
  Course introducing Python, C/C++, data analysis, large language model applications, and AI-agent development for scientific workflows.

Outreach activities

Selected outreach activities include:

- "Create a universe with a large accelerator - search for new physics beyond the Standard Model" [Link],
  Science cafe jointly hosted by ICEHAP-KMI, 10/2021.
- "Operation status of SuperKEKB/Belle II", KEK Press Conference, 08/2021.
- "Approaching mysteries of the universe with accelerator", The 1st KMI-ITbM Mix Cafe, 05/2021.
- "Introduction to particle physics", Shibuya Makuhari high school, 10/2020.
- "Come on B-mesons!" [Link], KEK open house, 09/2020.
- "Let's do science with Haiku", NHK E-television, 08/2020.

Selected Presentations since 2024

[1] "Real-Time Computing for Particle Physics: From AI on FPGA to AI Agents",
ICS 690 Seminar, University of Hawai'i at Mānoa, HI, US, 09/04/2026. Flyer

[2] "AI/ML on Hardware",
Belle II Summer Workshop, University of Louisville, KY, US, 07/2026.

[3] "Machine Detector Interface",
Belle II Summer Workshop, University of Louisville, KY, US, 07/2026.

[4] "The Belle II Upgrade Program",
Violation of Fundamental Symmetries with B mesons Workshop, Fermilab, IL, US, 2025.

[5] "AI/ML-Based Beam Diagnostics",
CEF Workshop: Versal and AI in FEE, KEK, Japan, 2025.

[6] "Front-end AI/ML Opportunities at SuperKEKB/Belle II",
CEF Workshop: Versal and AI in FEE, KEK, Japan, 2025.

[7] "Modern Electronics Education with AI/ML and FPGA",
CPAD2025 Workshop, University of Pennsylvania, PA, US, 2025.

[8] "Introduction to SuperKEKB/Belle II Machine Detector Interface",
Belle II Summer Workshop, Virginia Tech, VA, US, 2025.

[9] "Modern Electronics for Physicists: Project-Based Learning in PHYS476",
US-Japan Symposium, Sapporo, Japan, 05/2025.

[10] "Development and Implementation of Advanced Beam Diagnostic and Abort Systems in SuperKEKB",
US-Japan Symposium, Sapporo, Japan, 05/2025.

[11] "Modern Electronics for Physicists: Project-Based Learning in PHYS476",
ML4FE workshop, HI, US, 05/2025.

[12] "Current Status and Prospects of B-Physics",
KMI Symposium 2025, Nagoya, Japan, 03/2025.

[13] "Belle II Loss Monitor Group's Activities for Understanding the SBL Mechanism",
eeFACT conference, Tsukuba, Japan, 03/2025.

[14] "The Belle II Upgrade Program",
CPAD workshop 2024, TN, US, 11/2024.

Publications since 2024

Peer-reviewed Journals and Preprints

[1] H. Purwar *et al.*, "Migration of Belle II TOP Feature Extraction from the Zynq Processing System to PCIe40 Readout PCs", arXiv:2608.27814 (2026), submitted to *Nucl. Instrum. Methods Phys. Res. A*. arXiv

[2] Y.-T. Lai, T. Koga, Y. Nakazawa, N. Taniguchi, and K. Yoshihara, "Machine-Learning-Based Waveform Discrimination in the Front-End Electronics of the Belle II Central Drift Chamber for Cross-Talk Noise Reduction", arXiv:2607.25174 (2026), submitted to *IEEE Trans. Nucl. Sci.*. arXiv

[3] M. Andrew *et al.*, "Development of a Silicon-Based Ultra-Fast X-Ray Beam Size Monitor for SuperKEKB", arXiv:2607.08205 (2026), submitted to *Nucl. Instrum. Methods Phys. Res. A*. arXiv

[4] J. Gonski *et al.*, "Machine Learning on Heterogeneous, Edge, and Quantum Hardware for Particle Physics", *PRX Intelligence*, accepted 31 July 2026. DOI

[5] C. Ketter *et al.*, "Design and Commissioning of Readout Electronics for a K_L^0 and μ Detector at the Belle II Experiment", *Nucl. Instrum. Methods Phys. Res. A* **1082** (2026) 170893. DOI

[6] H. Atmacan *et al.*, "The Imaging Time-of-Propagation Detector at Belle II", *Nucl. Instrum. Methods Phys. Res. A* **1080** (2025) 170627. DOI

[7] K. Yoshihara *et al.*, "Development and Implementation of Advanced Beam Diagnostic and Abort Systems in SuperKEKB", *Nucl. Instrum. Methods Phys. Res. A* **1072** (2025) 170117. DOI

Conference Proceedings

[1] R. Nomaru *et al.*, "Development and First Beam Observations of an Ultra-Fast Bunch-by-Bunch X-Ray Beam Size Monitor at SuperKEKB", in *Proc. IPAC'26*, Deauville, France, 2026, paper MOV6603. DOI

[2] R. Nomaru *et al.*, "R&D of an Ultrafast X-Ray Beam Size Monitor for SuperKEKB", in *Proc. IBIC'25*, Liverpool, UK, 2025, pp. 435–438. DOI

[3] R. Zhang *et al.*, "Study on a Novel Laser Fast Abort System for SuperKEKB", in *Proc. IPAC'25*, Taipei, Taiwan, 2025, pp. 3048–3050. DOI

[4] H. Kaji *et al.*, "Upgrade of Beam Abort System at the SuperKEKB Positron Ring", in *Proc. IPAC'25*, Taipei, Taiwan, 2025, pp. 3037–3039. DOI

[5] K. Yoshihara *et al.*, "Loss Monitor Group's Activities for Understanding the SBL Mechanism at SuperKEKB/Belle II", in *Proc. eeFACT'25*, Tsukuba, Japan, 2025.